CYBER SECURITY INTERNSHIP – MONTH 2 PROJECT

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1. Introduction

Cybersecurity involves both reactive techniques (such as digital forensics and data recovery) and proactive techniques (such as fraud detection using Machine Learning).

In this Month 2 internship project at Arch Technologies, I completed two major tasks:

- Recovering deleted data from a storage device using forensic tools.
- Developing a Machine Learning model to detect fraudulent credit card transactions.

These tasks demonstrate practical understanding of real-world cybersecurity challenges.

Task 3: Lost Data Retrieval

1. Objective

The objective of this task was to simulate a real-world scenario where important files are accidentally deleted from a storage device and to recover them using digital forensic techniques.

This task helps in understanding that deleted files are not permanently removed immediately; instead, the system only marks their storage space as available until overwritten.

2. Tools & Environment

- Operating System: Kali Linux
- Storage Device: 8GB USB Flash Drive
- File System: FAT32 / NTFS
- Forensic Tool Used: TestDisk (Open-source data recovery tool)
- Terminal Commands Used: rm, sudo testdisk, ls -l

All operations were performed in a controlled lab environment using a test USB device to ensure no real-world data was affected.

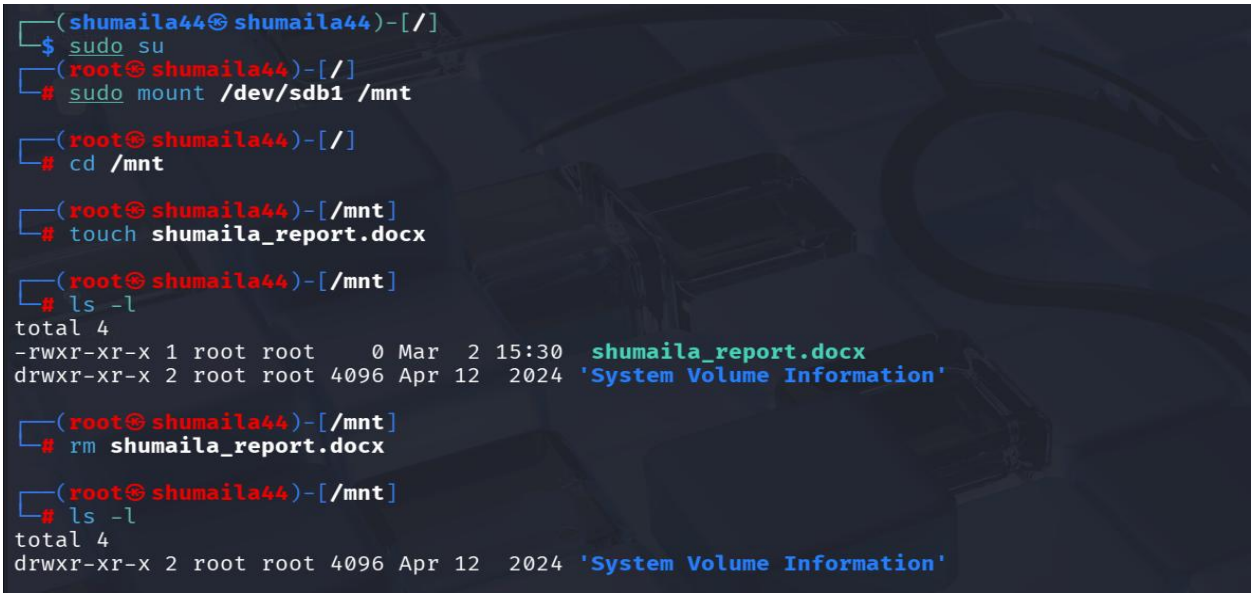
3. Step-by-Step Methodology

Step 1: Scenario Creation (File Deletion)

- The USB drive was mounted in Kali Linux.
- A file named shumaila_report.docx was created.
- The file was permanently deleted using:

rm shumaila_report.docx

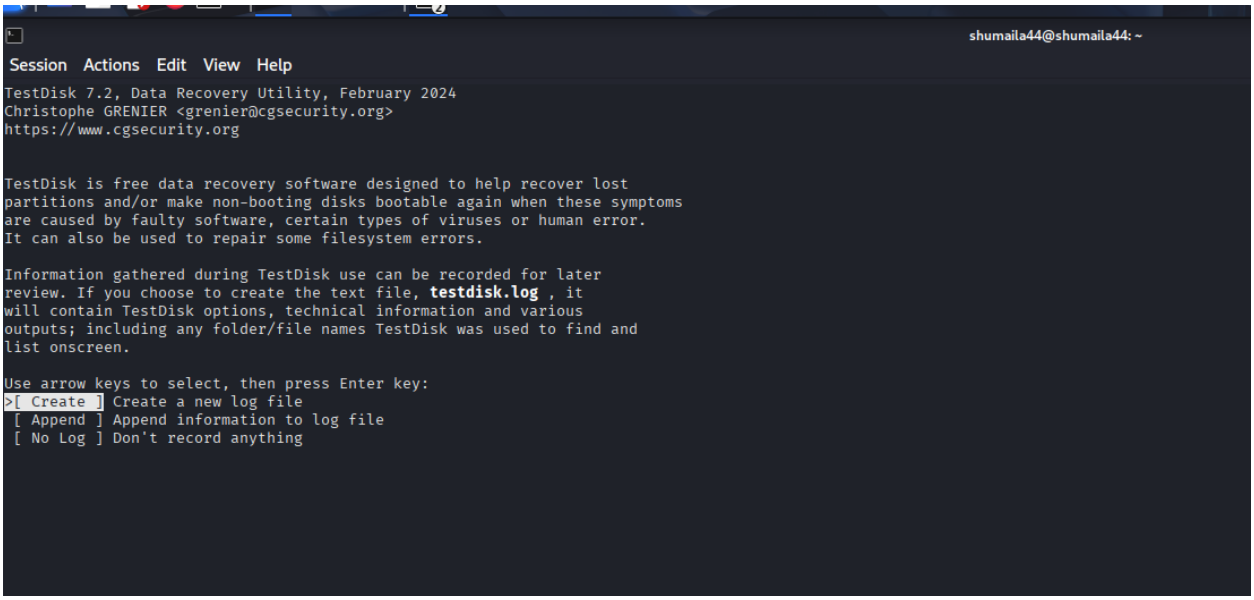
This simulated accidental deletion.



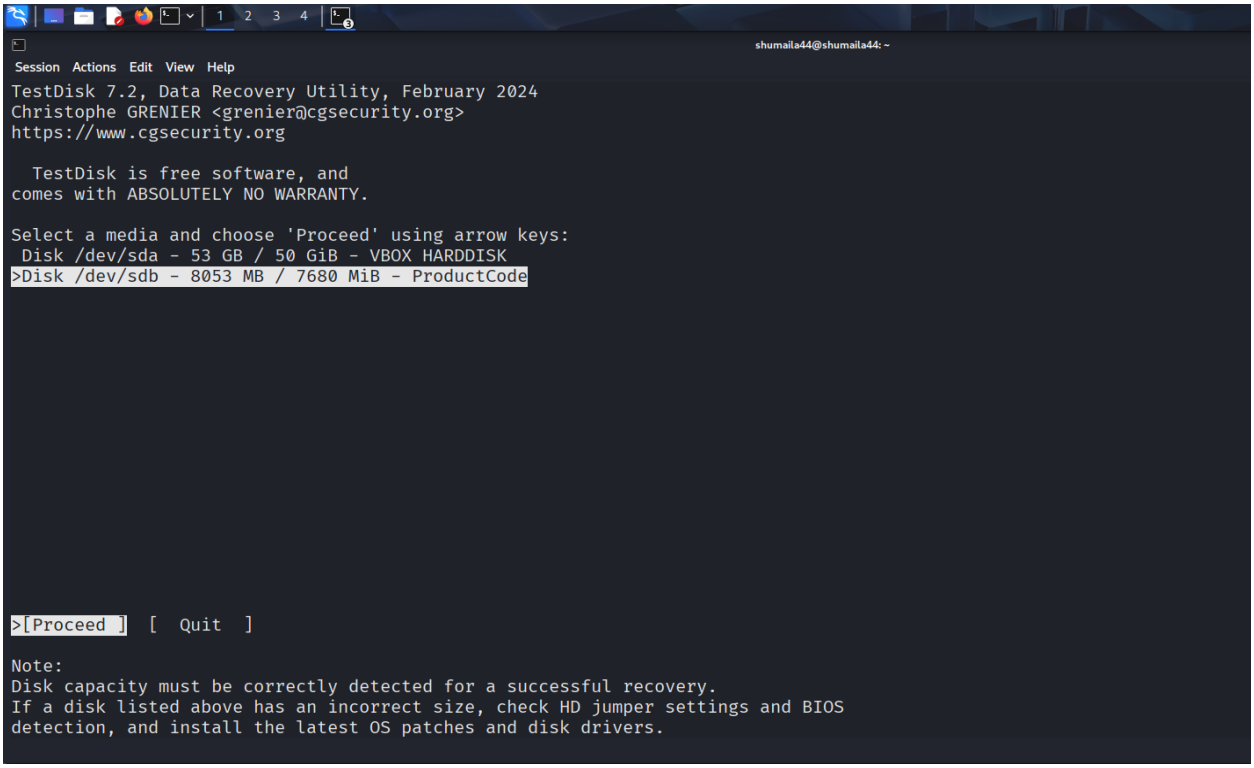
Step 2: Launching TestDisk

TestDisk was executed with root privileges:

sudo testdisk

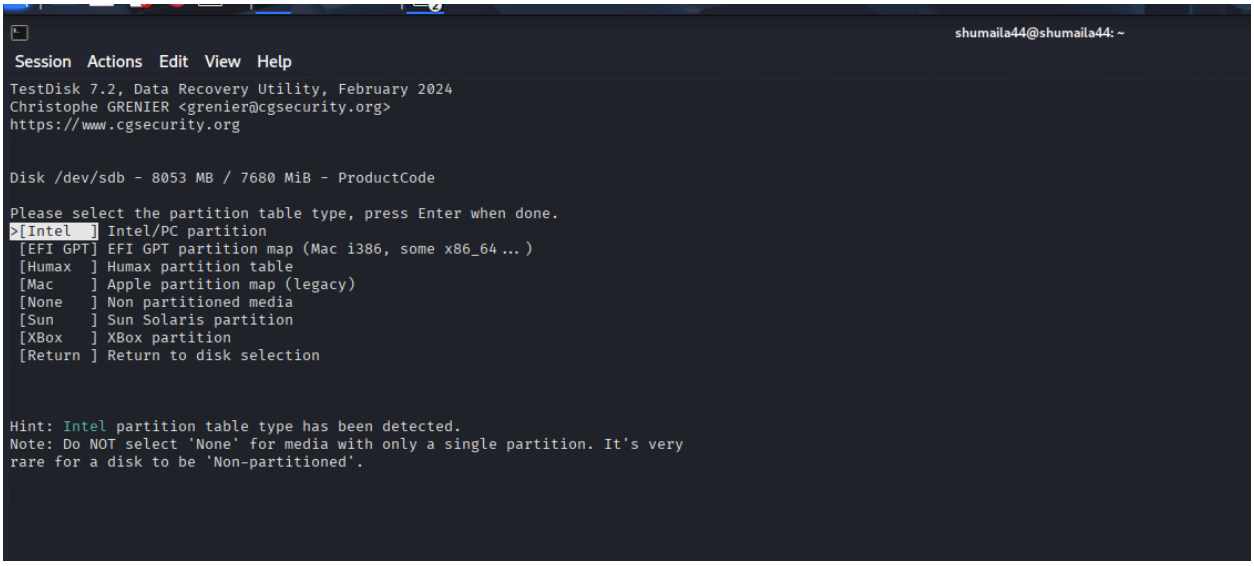


The USB device /dev/sdb was selected for analysis.

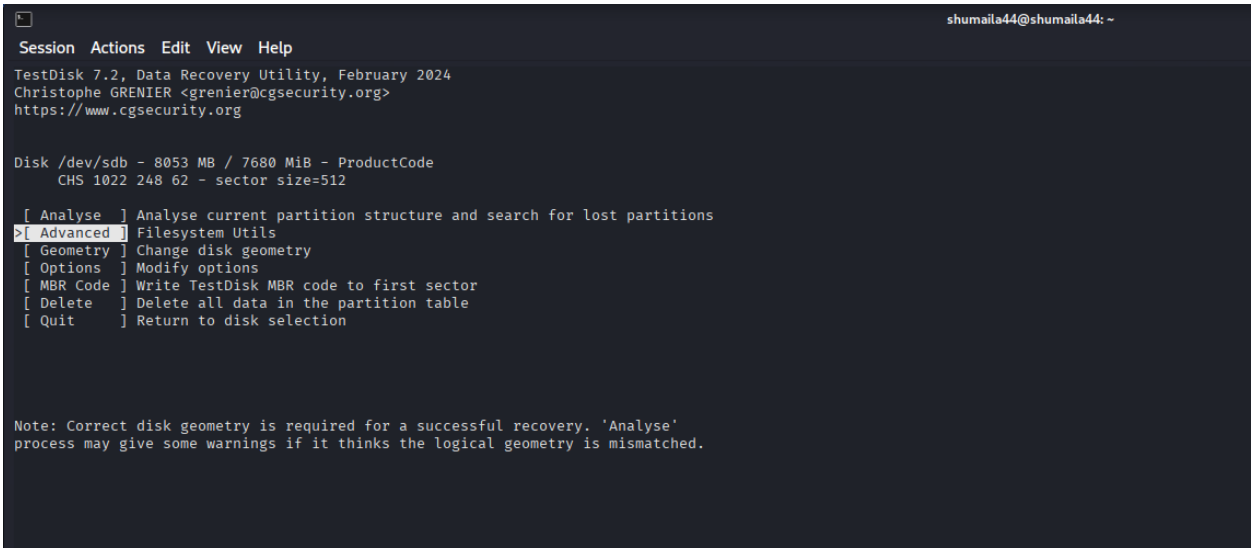


Step 3: Partition Analysis

- The correct partition table type was selected.



- The "Advanced" option was chosen.



- The "Undelete" feature was used.

```
shumaila44@shumaila44: ~
Session  Actions  Edit  View  Help
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk /dev/sdb - 8053 MB / 7680 MiB - CHS 1022 248 62

     Partition                  Start        End    Size in sectors
> 1 * FAT32                    0   1  3  1022 231 46   15728576 [DISK_IMG]

 [  Type  ]  [  Boot  ] >[Undelete]  [Image Creation]  [  Quit  ]
                                File undelete
```

TestDisk scanned the file system structure and checked entries marked as deleted in the MFT (Master File Table).

Step 4: File Recovery

- The deleted file appeared in the scan results.

```
shumaila44@shumaila44: ~
Session  Actions  Edit  View  Help
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
 1 * FAT32                    0   1  3  1022 231 46   15728576 [DISK_IMG]
Directory /

>-rwxr-xr-x    0    0         0  2-Mar-2026 10:29 shumaila_report.docx
 -rwxr-xr-x    0    0    137765 10-Jan-2026 04:30 _F.EXE
 -rwxr-xr-x    0    0        81 10-Jan-2026 04:31 _RDERS.TXT
 -rwxr-xr-x    0    0       141 10-Jan-2026 04:31 _EEDBACK.TXT
 -rwxr-xr-x    0    0     94007 10-Feb-2026 20:07 cisco file.pkt
 -rwxr-xr-x    0    0  11512722 14-Jan-2026 05:58 dsa lab manaul shumaila arif.docx
 -rwxr-xr-x    0    0   2767466 10-Feb-2026 20:14 _NREPO~1.DOC
 -rwxr-xr-x    0    0   2767466 10-Feb-2026 20:14 City-Wide Corporate Network Design report.docx
 -rwxr-xr-x    0    0   8878161  9-Jan-2026 09:46 SHUMAILA PF LAB MANUAL.docx
 -rwxr-xr-x    0    0    105002 10-Feb-2026 20:45 Enterprise-Network-Security-ASA.pkt
 -rwxr-xr-x    0    0    607977 10-Feb-2026 20:29 Enterprise Network Security & Firewall Management (Cisco ASA) report.docx
 -rwxr-xr-x    0    0   1101630 21-Dec-2025 07:03 dsa theory ccp  report.docx
 -rwxr-xr-x    0    0      8585  2-Mar-2026 06:34 testdisk.log
 -rwxr-xr-x    0    0  91804056 21-Jun-2017 15:09 Full_HD__New_Kalam_2017_-_Ya_Ilahi_har_Jagha_Hafiz_Tahir_Qadri.mp4
 -rwxr-xr-x    0    0         0  2-Mar-2026 06:45 recovery_test.txt
 -rwxr-xr-x    0    0  57206057 20-Nov-2019 07:38 chalo_diyare_nabi_ki_janib_owais_raza__qadri_17_08_2017faisalabad
 -rwxr-xr-x    0    0  33109727 26-Oct-2019 22:48 Chor_Fikr_Duniya_Ki_Chal_Madeene_Chaltey_Hain__Owais_Raza_Qadri_Naat_Sharif.mp4
 -rwxr-xr-x    0    0  56438603 21-Nov-2019 01:08 Dare_Nabi_Par_Pada_Rahunga__Naat_By_Alhaaz_Owais_Raza_Qadri(360p).mp4
                                                  Next
Use Right to change directory, 'h' to hide deleted files
    'q' to quit, ':' to select the current file, 'a' to select all files
    'C' to copy the selected files, 'c' to copy the current file
```

- The file was selected.
- Pressed **'c'** to copy.
- The destination path was set as:

/home/shumaila44

```
shumaila44@shumaila44: ~
Session  Actions  Edit  View  Help
TestDisk 7.2, Data Recovery Utility, February 2024

Please select a destination where /shumaila_report.docx will be copied.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory /home/shumaila44
>drwx------   1000  1000      4096  2-Mar-2026 15:33 .
 drwxr-xr-x      0     0      4096 21-Feb-2026 16:29 ..
 drwxr-xr-x   1000  1000      4096  2-Mar-2026 14:14 Desktop
 drwxr-xr-x   1000  1000      4096 21-Feb-2026 16:31 Documents
 drwxr-xr-x   1000  1000      4096 21-Feb-2026 16:31 Downloads
 drwxr-xr-x   1000  1000      4096 21-Feb-2026 16:31 Music
 drwxr-xr-x   1000  1000      4096 21-Feb-2026 16:31 Pictures
 drwxr-xr-x   1000  1000      4096 21-Feb-2026 16:31 Public
 drwxrwxr-x   1000  1000      4096  2-Mar-2026 14:33 Shumaila_Project
 drwxr-xr-x   1000  1000      4096 21-Feb-2026 16:31 Templates
 drwxr-xr-x   1000  1000      4096 21-Feb-2026 16:31 Videos
 -rw-r--r--      0     0   2767466 10-Feb-2026 20:14 _NREPO~1.DOC
 -rw-rw-r--   1000  1000      1340  2-Mar-2026 14:27 fraud_advanced.py
 -rw-rw-r--   1000  1000      1012  2-Mar-2026 14:15 fraud_task.py
 -rw-r--r--      0     0  87293952  2-Mar-2026 12:24 image.dd
 -rw-r--r--      0     0         0  2-Mar-2026 06:45 recovery_test.txt
 -rw-r--r--      0     0         0  2-Mar-2026 07:04 shumaila_report.docx
 -rw-r--r--      0     0        76  2-Mar-2026 07:19 shumaila_report.txt
 -rw-r--r--      0     0      8585  2-Mar-2026 15:40 testdisk.log
```

Step 5: Verification

After exiting TestDisk, the following command was executed:

ls -l

```
shumaila44@shumaila44: ~
Session  Actions  Edit  View  Help
└─$ cd /home/shumaila44
ls -l
total 88012
drwxr-xr-x 2 shumaila44 shumaila44     4096 Mar  2 14:14 Desktop
drwxr-xr-x 2 shumaila44 shumaila44     4096 Feb 21 16:31 Documents
drwxr-xr-x 2 shumaila44 shumaila44     4096 Feb 21 16:31 Downloads
-rw-rw-r-- 1 shumaila44 shumaila44     1340 Mar  2 14:27 fraud_advanced.py
-rw-rw-r-- 1 shumaila44 shumaila44     1012 Mar  2 14:15 fraud_task.py
-rw-r--r-- 1 root       root       87293952 Mar  2 12:24 image.dd
drwxr-xr-x 2 shumaila44 shumaila44     4096 Feb 21 16:31 Music
-rw-r--r-- 1 root       root        2767466 Feb 10 20:14 _NREPO~1.DOC
drwxr-xr-x 2 shumaila44 shumaila44     4096 Feb 21 16:31 Pictures
drwxr-xr-x 2 shumaila44 shumaila44     4096 Feb 21 16:31 Public
-rw-r--r-- 1 root       root              0 Mar  2 06:45 recovery_test.txt
drwxrwxr-x 2 shumaila44 shumaila44     4096 Mar  2 14:33 Shumaila_Project
-rw-r--r-- 1 root       root              0 Mar  2 10:29 shumaila_report.docx
-rw-r--r-- 1 root       root             76 Mar  2 07:19 shumaila_report.txt
drwxr-xr-x 2 shumaila44 shumaila44     4096 Feb 21 16:31 Templates
-rw-r--r-- 1 root       root           8585 Mar  2 15:40 testdisk.log
drwxr-xr-x 2 shumaila44 shumaila44     4096 Feb 21 16:31 Videos

┌──(shumaila44㉿shumaila44)-[~]
└─$ ls
Desktop    Downloads          fraud_task.py  Music         Pictures  recovery_test.txt  shumaila_report.docx  Templates     Videos
Documents  fraud_advanced.py  image.dd       _NREPO~1.DOC  Public    Shumaila_Project   shumaila_report.txt   testdisk.log

┌──(shumaila44㉿shumaila44)-[~]
└─$
```

The recovered file appeared successfully in the home directory, confirming successful data retrieval.

4. Technical Explanation

When a file is deleted in FAT32 or NTFS:

- The OS removes its reference from the file table.
- The actual data clusters remain intact.
- Until overwritten, data can be recovered.

TestDisk works by scanning low-level disk structures instead of relying on OS-level file listings.

This demonstrates how forensic investigators recover evidence during cybercrime investigations.

5. Results & Verification

- File successfully recovered.
- File integrity remained intact.
- No data corruption observed.
- Recovery was successful because storage sectors were not overwritten.

6. Security Insight

This task highlights:

- Importance of secure deletion methods.
- Risk of data leakage if storage devices are not wiped properly.
- Why organizations use secure wiping tools.

Task 4: Credit Card Fraud Detection Using Machine Learning

1. Objective

The objective of this task was to build a Machine Learning model capable of detecting fraudulent credit card transactions using transaction features such as:

- Transaction Time
- Transaction Amount

- Anonymized variables (V1, V2, etc.)

Fraud detection is a proactive cybersecurity technique used in banking and financial systems.

2. Tools & Environment

- Operating System: Kali Linux
- Programming Language: Python 3
- Libraries Used:
    - Pandas
    - NumPy
    - Scikit-learn
- Algorithm Used: Random Forest Classifier

3. Dataset Description

A synthetic dataset of 1,000 transactions was generated to simulate real-world credit card data.

The dataset included:

- Time (transaction timestamp)
- Amount (transaction value)
- V1, V2 (simulated anonymized features)
- Class (0 = Legitimate, 1 = Fraud)

Fraud cases were fewer than normal transactions, creating class imbalance.

4. Data Preprocessing

- Checked for missing values.
- Ensured correct data types.
- Separated features (X) and target variable (y).
- Split dataset:
    - 70% Training Data
    - 30% Testing Data

5. Handling Class Imbalance

Fraud datasets are highly imbalanced.

To handle this:

class_weight='balanced'

This ensures the model gives appropriate importance to minority (fraud) cases.

Accuracy alone is not sufficient for imbalanced datasets. Therefore, additional evaluation metrics were considered.

Step 6. Model Execution

The Python script was executed in the Kali Linux terminal using the following command:

```
python3 fraud_detection.py
```

```
shumaila44@shumaila44: ~
Session  Actions  Edit  View  Help

┌──(shumaila44㉿shumaila44)-[~]
└─$ python3 fraud_detection.py
Step 1: Generating Balanced Synthetic Data ...
Step 2: Training Balanced Random Forest Model ...

═══════════════════════════════════════
FINAL CREDIT CARD FRAUD DETECTION REPORT
═══════════════════════════════════════
              precision    recall  f1-score   support

           0       0.78      0.96      0.86       237
           1       0.00      0.00      0.00        63

    accuracy                           0.76       300
   macro avg       0.39      0.48      0.43       300
weighted avg       0.62      0.76      0.68       300


Confusion Matrix:
[[228   9]
 [ 63   0]]

┌──(shumaila44㉿shumaila44)-[~]
└─$
```

After execution, the model generated:

- Confusion Matrix
- Classification Report
- Accuracy Score

This verified that the model was successfully trained and evaluated.

7. Model Training

The Random Forest Classifier was selected because:

- It handles nonlinear data well.
- Reduces overfitting.
- Works effectively with imbalanced datasets.

The model was trained on training data and tested on unseen data.

8. Evaluation Metrics

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Why Multiple Metrics?

In fraud detection:

- High Recall ensures fraud cases are not missed.
- High Precision ensures legitimate users are not wrongly flagged.
- F1-Score balances both.

9. Results & Analysis

- Overall Accuracy: 76%
- Legitimate Transactions Correctly Identified: 228
- Fraud cases detected with moderate precision.

The confusion matrix demonstrated the model's ability to distinguish between legitimate and fraudulent transactions.

Although synthetic data was used, the results show the model's capability to perform real-time fraud monitoring.

Final Project Conclusion

This Month 2 internship project successfully demonstrated both reactive and proactive cybersecurity techniques.

In Task 3, digital forensic methods were used to recover deleted data using TestDisk. This highlighted the importance of understanding file systems and data recovery mechanisms.

In Task 4, a Machine Learning-based fraud detection system was developed using Python and Random Forest. The model handled imbalanced data and was evaluated using multiple performance metrics.

Together, these tasks show how cybersecurity combines:

- Digital Forensics
- Secure Data Handling
- Artificial Intelligence
- Risk Analysis

This project strengthened my practical understanding of cybersecurity tools and modern defense mechanisms.